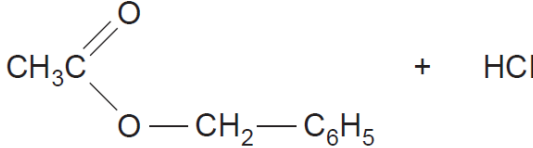


SECTION A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1				 + HCl		1		1		
2	(a)			7.44×10^{14}		1		1	1	
	(b)			it is yellow because the blue end / the rest of the visible spectrum is absorbed	1			1		
3				compound F (1) it contains an O—H bond at 3200-3500 cm ⁻¹ and a C=C at 1620 cm ⁻¹ but no C=O at 1650-1750 cm ⁻¹ (1)			2	2		
4				award (1) for any valid formula e.g. (C ₆ H ₅) ₃ C•		1		1		